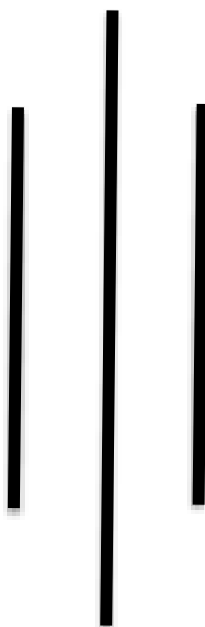




Tribhuvan University

Institute of Science and Technology

Amrit Campus



A Lab Report on
Simulation and Modelling (CSC-328)

Submitted to:
Department of Computer Science & Information Technology
Amrit Campus, Thamel, Kathmandu
Tribhuvan University

In partial fulfillment of the requirement for the *Bachelor's Degree in*
Computer Science & Information Technology

Submitted By:

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Submission Date:

Submitted To:

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Signature

LABWORKS

S.No.	Name of Experiments	Date of Submission	Signature	Remarks
	<u>C Programs</u>			
1.	Write program to implement Monte Carlo's simulation.			
2.	Write program to implement queuing system and calculate different queue parameters.			
3.	Write program to implement Markov chain.			
4.	Write program to generate random number using linear congruential method.			
5.	Write program to generate random number using multiplicative congruential method.			
6.	Write program to generate random number using mid square method.			
7.	Write program to implement KS-test.			
8.	Write program to implement autocorrelation test.			
9.	Write program to implement Chi-Square test.			
10.	Write program to implement gap test.			
	<u>GPSS Programs</u>			
11.	Write a GPSS program to simulate the Joe's Barber Shop, where customers arrive every 18 ± 3 minutes and are served by a single barber named JOE with service time 10 ± 3 minutes for each customer.			
12.	A machine tool in a manufacturing shop produces parts at the rate of one part every 5 minutes. As they finish, the parts go to an inspector who takes 4 ± 3 minutes to examine each one and rejects 10% of parts. Write a GPSS program to simulate this system.			
13.	Write the GPSS program for the same problem as LAB-12 using the multiple servers (3-servers) and modify the timing parameters.			
14.	Workers arrive at a supply store at a rate of 6 ± 2 minutes. Their requisitions are processed by one of two clerks, each taking 8 ± 2 minutes to process a requisition. The processed requisitions are then passed to a single storekeeper, who fills them one at a time, taking 6 ± 3 minutes for each. Write a GPSS program and draw the block diagram.			
15.	Parts are produced at a rate of every 10 minutes. The parts are of two types: A and B, mixed			

	randomly, with about 10 % being type B. A separate inspector is assigned to examine each type of part. Both inspectors reject 10% of the parts they inspect. Write a GPSS program for 1000 items. (Inspector A takes 6 ± 2 minutes to examine and inspector B takes 10 ± 2 minutes)			
16.	Write a GPSS program and draw the block diagram to simulate 1000 vehicles arriving at fuel station: - Fuel station has 2 fuel nozzles. Vehicles arrive at fuel station at an interval of 3 ± 2 minutes. The fueling process for each vehicle takes 5 ± 2 minutes.			